

Table 2-4 Functions of three variables

<u>x</u>	<u>y</u>	<u>z</u>	<u>Function f_1</u>	<u>Function f_2</u>
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

• A Boolean function may be expressed algebraically from a given truth table by forming a minterm for each combination of the variables which produces a 1 in the function, and then taking the OR of all these terms.

$$f_1 = x'y'z + xy'z + xyz = m_1 + m_4 + m_7$$

$$f_2 = x'yz + xy'z + xyz = m_3 + m_5 + m_6 + m_7$$

§ Important property of Boolean algebra

Any Boolean function can be expressed as a sum of minterms: → (by "sum" is meant the ORing of terms).

• The complement of f_1 is read as: $f_1' = x'y'z' + x'yz' + x'y'z + xy'z' + xyz'$

• If we take the complement of f_1 , we obtain the function f_1' :

$$f_1' = (x+y+z) \cdot (x+y'+z) \cdot (x+y'+z') \cdot (x'+y+z) \cdot (x'+y'+z)$$

$$= M_0 \cdot M_2 \cdot M_3 \cdot M_5 \cdot M_6$$

• Similarly, it is possible to read the expression for f_2 from the table;

$$f_2 = (x+y+z) \cdot (x+y+z') \cdot (x+y'+z) \cdot (x'+y+z)$$

$$= M_0 \cdot M_1 \cdot M_2 \cdot M_4$$

Second important property of Boolean algebra

Any Boolean function can be expressed as a product of maxterms (by "product" is meant the ANDing of terms).